Student Performance Analysis

~Exploratory Data Analysis (EDA) Summary Report

Prepared By:

Mohammed Nayeema

Objective

The objective of this project was to analyze student exam performance using Exploratory Data Analysis (EDA). The analysis was carried out to understand the dataset, perform data cleaning, calculate statistical measures, detect outliers, analyze correlations, and visualize the data to identify meaningful patterns in student performance.

Dataset Information

Dataset Name: *Students Performance Dataset*

Source: *Kaggle*

Dataset Link:

https://www.kaggle.com/datasets/spscientist/students-performance-in-exams

Dataset Description

The dataset contains information about students and their academic performance. It includes demographic details, parental education, lunch type, test preparation course, and exam scores in Mathematics, Reading, and Writing.

Data Exploration

The dataset was first explored to understand its structure and contents. Functions such as head(), shape, info(), and describe() were used to examine the dataset. This helped identify the number of records, column names, data types, and basic statistical information before starting the analysis.

Data Cleaning

The dataset was checked for missing values and duplicate records. Missing values were handled appropriately, and duplicate entries were removed to improve the quality of the data. These preprocessing steps ensured that the dataset was clean and suitable for further analysis.

Statistical Analysis

Descriptive statistical measures including mean, median, mode, variance, and standard deviation were calculated for the numerical columns. These measures provided a better understanding of the distribution, central tendency, and variability of student scores.

Outlier Detection

Outliers were detected using the Interquartile Range (IQR) method. Box plots were created to visualize extreme values in the dataset. Identifying outliers helped in understanding unusual observations that could affect the overall analysis.

Correlation Analysis

Correlation analysis was performed to study the relationship between numerical variables. A correlation heatmap was generated to visualize the strength of relationships among Mathematics, Reading, and Writing scores. The analysis showed a strong positive relationship between Reading and Writing scores.

Data Visualization

Different charts were created to better understand the dataset and its patterns. These included Bar Chart, Line Chart, Histogram, Scatter Plot, Correlation Heatmap, and Box Plot. The visualizations helped in understanding score distributions and relationships between different variables.

Key Insights

The dataset was successfully cleaned before analysis. Statistical analysis provided useful information about the distribution of student scores. Outliers were identified using the IQR method, and correlation analysis showed that Reading and Writing scores were strongly related. The visualizations made it easier to identify patterns and compare student performance across different features.

Conclusion

This project provided practical experience in performing Exploratory Data Analysis using Python. It covered important concepts such as data cleaning, descriptive statistics, outlier detection, correlation analysis, and data visualization. The analysis helped in gaining meaningful insights into student performance and demonstrated how EDA can be used to understand real-world datasets.

11. Tools Used

- *Python*
- *Pandas*
- *NumPy*
- *Matplotlib*
- *Seaborn*
- *Google Colab*

12. Project Outcome

Through this project, I gained hands-on experience in:

- *Data Cleaning*
- *Exploratory Data Analysis (EDA)*
- *Statistical Analysis*
- *Correlation Analysis*
- *Outlier Detection*
- *Data Visualization*
- *Interpreting analytical result*